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THE PROCESSES AT THE MERCURY DROPPING CATHODE.

PART I. THE DEPOSITION OF METALS.

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Introduction.

The methods hitherto employed of following electrode potentials during polarisations contrast markedly—as regard exactness—with the E.M.F. determinations of primary galvanic cells; this is due to the circumstance that for polarisations more or less considerable current densities are used, whereas galvanic cells are studied in a "currentless" condition. Thus the polarisation "current-voltage" diagrams bear the character of a technical research rather than of exactly reproducible measurements, which would show strict thermodynamic relationships.

To approach nearer to the stage of reversibility in investigating the polarisation phenomena the author has applied [1] to the study of electrolytic processes a dropping mercury cathode as used by B. Kučera in the determination of the interfacial tension of polarised mercury.[2]

Fig. 1 shows a suitable form of the electrolytic vessel.

Through a rubber or cork stopper A, placed at the top of a long-necked conical flask so as not to contaminate the solutions, the mercury cathode B is inserted, consisting of a thick-walled capillary, the lower end of which is drawn into a tip allowing mercury to drop out slowly. The upper end of the capillary is connected by a rubber tubing, ca. 50 cms. long, to a mercury reservoir, by the level of which the rate of dropping can be regulated; the usual drop time was ca. 3 seconds.

In order to expel air and remove oxidising impurities from the solution in the vessel pure hydrogen is let in before electrolysis through the glass tube D, it passes the solution and escapes through the syphon C. The large bottom mercury layer Hg serves as anode, from which a platinum contact leads to the connection in F.

Through the tap H the solution or mercury may be let out, and through E fresh mercury can be poured in, if required.

When the solution is properly reduced by hydrogen, which is effected after two or three hours of slow passage, the siphon C is lowered and filled with the solution, thus establishing a connection with a normal calomel electrode, to which the potential of the bottom mercury layer is referred.

[1] *Phil. Mag.*, **45** (1923), p. 303.

[2] *Drud. Ann.*, **11** (1903), p. 529.

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The negative pole of the polarising source is then connected to the mercury reservoir, making the drops cathode, whilst the mercury layer is joined with the positive pole; the polarising E.M.F. is then slowly increased and the current which passes through the solution is determined by a sensitive galvanometer G (a d'Arsonval with 4 mm. deflection corresponding to 10^{-8} amp.).

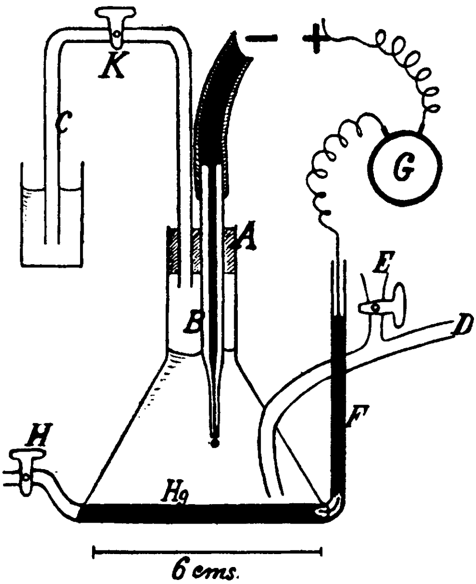

FIG. 1.

The constant renewal of mercury surface at the dropping cathode, which prevents "concentration-polarisation" and hydrogen evolution, effects slight fluctuations due to the charge acquired by the spreading of the mercury surface (the electrocapillary phenomenon), which causes the galvanometer mirror to oscillate 1-2 millimeters of the scale-division, corresponding to changes in intensity of $2 - 5 \times 10^{-9}$ amp.; the current which passes the cell before electrolysis of the solution sets in is of the order of $10^{-8} - 10^{-7}$ amp.

Electrodeposition in Mercury.

If a properly reduced solution be polarised, the current must all be due to the ionic deposition in the mercury drop; it is therefore important to avoid in the solution traces of mercury salts, which would deposit already at very small polarisations. Even very sparingly soluble salts such as mercurous chloride or mercuric oxide cause by their presence a large increase of current; this is especially marked in more concentrated (ca. normal) solutions of alkali hydroxides or chlorides, showing that these solutions then contain mercury in complex ions. Solutions of bromides and iodides, when standing longer over mercury, show the presence of such complexes so strongly, that only concentrations from decinormal downwards can be used

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for electrolysis in this method. A trace of cupric hydroxide when added to an alkaline solution brings the galvanometer mirror off the scale even at 0·2 – 0·3 volt polarisation.

If all such impurities of nobler metals are absent, the current passing the solution is very small and the oscillations of the mirror are slight, vanishing entirely at the potential at which the interfacial tension of mercury in the solution is maximal (*i.e.*, ca. – 0·56 volt from the calomel electrode); this, then, tests the purity of the solution and furnishes a check for the right adjustment of the arrangement.

Fig. 2 shows some of the characteristic "current voltage" curves as obtained by this method. All measurements were carried out at room temperature varying between 18°-20° C.

During polarisation ions are discharged at the drops, causing thereby a back E.M.F. equal and opposite to the polarising force; the current due to this deposition is minute, $10^{-8} - 10^{-7}$ amp. per sq. mm., as will be observed on the horizontal part of the curves. However, at a certain potential of the polarised drop, termed the "deposition potential," and denoted on the diagrams by arrows, a sudden increase of current begins.

Three processes might cause this increase, provided oxidising agents are absent and the ions in the solution are at their lowest stage of oxidation:—

(1) Combination with mercury and diffusion into the drops.

(2) A new phase formation at the mercury drop surface.

(3) Diffusion of a deposited volatile product into the surrounding solution.

The first case occurs with all metals which possess chemical affinity for mercury, *i.e.*, form amalgams; their deposition potential into mercury, π_N, is therefore much more positive than the electrolytic potential E.P. of the pure metal, the difference of these two potentials being a measure of the free energy A of amalgam formation.

As this has been discussed already in the previous communication (l.c. *Phil. Mag.*), only some recent values are added here:—

The metal	E.P.	π_N	$A = \pi_N -$ E.P.
Zn	– 1·043	– ·865	·178
Cd	– ·683	– ·348	·335
Pb	– ·415	– ·264	·151

Again, the affinity is expressed in volt-faradays (per gram equivalent). The values previously derived for alkali metals are about 1 volt-faraday (23,000 cals.) larger; this agrees with the fact that alkali metals combine strongly with mercury, whereas zinc and lead dissolve simply; the cadmium amalgamation, however, is of a more complicated nature, as is evident from the electrode potentials of such alloys.

The second case will occur if the atoms of the metal in solution have a limited solubility in mercury or are insoluble, *i.e.*, have no affinity for mercury. Here it must be borne in mind that the deposit of such a metal forms on the mercury surface by no means a coherent piece of a metallic lamella, held together by the interatomic bonds of crystalline structure, but must resemble more a concentrated metallic vapour, with a much greater vapour tension and activity than that of the metallic rod. Consequently the "solution tension" of this—atomic—form of the metal must be much greater and the electrolytic potential more negative than in the compact form.

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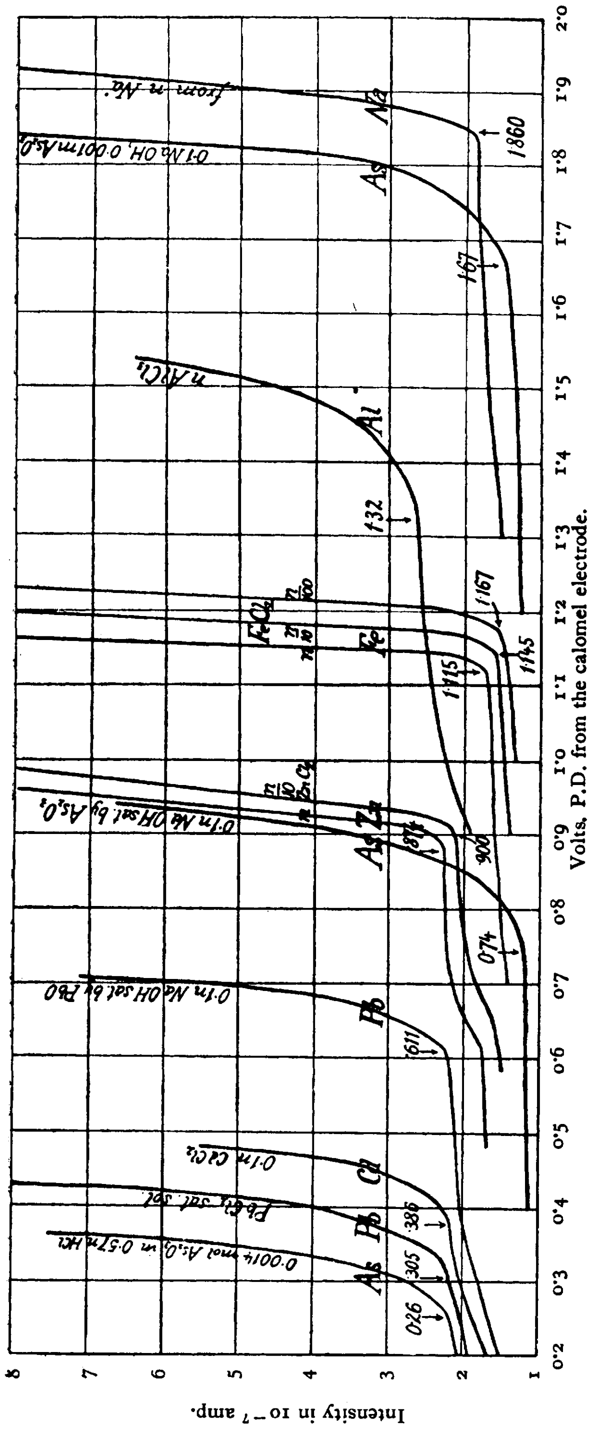

FIG. 2.

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In such cases we must expect the electro-deposition in mercury to take place at a potential even more negative than is the E.P. of the metallic rod (an instance of, as it is sometimes denoted, " cathodic passivity ").

This has been examined in the cases of iron and arsenic, which are known to possess little affinity for mercury.

To investigate the process of iron deposition at the dropping electrode, solutions of *n*, 0·1*n*, and 0·01*n* hydrochloric acid, and 0·1*n* hydrobromic acid, were saturated by dissolving in them (Merck's) pure iron in the form of filings, or piano-wire, in an atmosphere of hydrogen. Repeated electrolysis with solutions prepared at different times gave uniform results, *viz.* the deposition from $\frac{n}{1}$ $FeCl_2$ at − 1·115 *v.*, from $\frac{n}{10}$ $FeCl_2$ and $\frac{n}{10}$ $FeBr_2$ at − 1·145 *v.*, and from the most dilute ferrous solution, to which platinised platinum had to be added to facilitate the dissolution of iron and potassium chloride to increase the conductance. The deposition started at − 1·167 volt from the *n* calomel potential.

During these experiments it happened often that some of the iron filings fell upon the bottom mercury layer changing at once its potential from the ordinary calomel value to − 0·714 volt in $\frac{n}{1}$ $FeCl_2$, and to − ·744 in the 0·1*n* $FeCl_2$ or $FeBr_2$. This value remained constant during electrolysis, and must be regarded as the potential of pure iron in the corresponding solution of ferrous ions; it agrees with the values obtained by Richards and Behr[1] and by Foerster.[2] It is well known[3] that even on metallic iron ferrous ions are deposited at a potential much more negative than that corresponding to its electrolytic potential; however, the value was never found as negative as − 1·1 volt. The explanation of this "cathodic passivity" is to be sought in an alloy formation with hydrogen (see Foerster, *l.c.*); however, such a process seems improbable at the dropping cathode, since an increase of acidity of the ferrous solution, which would enrich the hydrogen deposition, had no marked effect upon the deposition potential of iron. The value − 1·115 *v.* (from the *n* calomel electrode) represents the deposition of iron forming a solution in mercury, whereas − 0·714 relates to the potential of metallic iron.

The exceptionally small vapour tension of iron and its correspondingly very great energy of evaporation—which is estimated by Grüneisen[4] to be 113 kg.-cals., *i.e.*, 2 volt-faradays per equivalent at absol. zero—and, again, the well-known unstability of iron amalgam enable us to understand, that the potential at which single atoms are deposited in mercury must be more negative than the potential of solid iron.

Applying the affinity relationship, we obtain for the affinity of metallic iron towards mercury a negative value—0·401 volt-faradays; this means that metallic iron does not combine with mercury and that the iron amalgam is metastable, *i.e.*, unstable when in contact with metallic iron. Only when in a state corresponding in activity to a much greater vapour tension can the reaction of iron with mercury proceed spontaneously; in fact, the amalgam can only be prepared through atomic dispersion, *e.g.*, by electrolysis, and is very unstable.

The case of arsenic is simpler, since it does not form an amalgam; the sudden increase of current should therefore appear, when an amorphous

[1] *Zeitsch. f. p. Chem.*, **58** (1907), p. 301.
[2] *Abhandl. Bunsen.-Ges.*, Nr. 2 (1909).
[3] Coffetti and Foerster, *Ber. d. Deutsch. chem. Ges.*, **38** (1905), p. 2940.
[4] *Verh. d. D. phys. Ges.*, **10** (1912), p. 327.

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layer of arsenic begins to be deposited. Since such a thin layer of arsenic must have a greater vapour tension than the solid crystal, the arsenic deposition potential in mercury should be more negative than the electrode potential of metallic arsenic in the same solution. The latter is, however, very unstable; measurements were carried out in that respect by Mr. Bayerle in this laboratory with a platinum wire upon which arsenic had been electrically deposited from arsenite solutions. The potential of this electrode, dipping into alkaline arsenite solutions in an atmosphere of hydrogen, is very variable in *n* sodium hydroxide solutions, but more stable in decinormal alkaline solutions. Thus in 0·095 *n* sodium hydroxide with 0·00945 mols. of As_2O_3 the potential fluctuates between −0·893 and −0·900; with 0·0312 mols. of arsenious oxide from −0·882 to −0·887 volt. In a decinormal hydrochloric acid with 0·02 mols. of As_2O_3, the value varied between +0·046 and +0·091*v*. The deposition potentials of arsenic are much more negative than these values; thus the affinity of arsenic for mercury again appears negative.

It will be noticed from the "current voltage" curves (in Fig. 2) that in the case of arsenic the turning points are indistinct and show too great shifts, which cannot be accounted for if the process which furnishes arsenious cations, $As^{\cdot\cdot\cdot}$, proceeds with great velocity; this suggests a retardation due to imperfect ionic splitting.

The third process which may occur at the polarised drop and is realised in the case of hydrogen deposition, will be—since closely connected with the overpotential phenomenon—discussed in Part II. of this communication.

Applications.

Ionic Equilibria.—The reversible ionic deposition at the mercury drops, which has been proved previously to proceed at potentials changing according to the formula $\frac{RT}{n}\log c$, where c denotes the concentration of metallic ions, and n their ionic charge, enables us to investigate by this method ionic equilibria. It is applicable also in cases when the potentials of ordinary metallic electrodes cease to be sensitive to the presence of their ions, as in cases of passivity or other solution actions on the surfaces, because the freshly deposited atoms in the mercury surface can never be passive nor are they attacked by the solution.

To show the sensitivity of the current towards ions, a diagram of "current voltage" curves is given in Fig. 3 on the electrolysis of lead solutions, which has been carried out by Miss H. Kadlcová, and from which various conclusions can be drawn.

The first curve shows the way the readings were followed, *viz.* by increasing in the critical region the voltage by 5 millivolts and thus finding the point at which decomposition begins. We observe that the change of the deposition potential of lead ions from a saturated solution of plumbous chloride to one diluted tenfold is 25 millivolts, just as found in concentration cells with a divalent cation. From the values we can deduce, taking into consideration the solubility of lead chloride, that the deposition potential of a solution normal in plumbous chloride would be −0·273 volts. The deposition potential of lead from a *n* sodium hydroxide solution saturated with yellow plumbous oxide is −0·669. We thus obtain for the

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solubility product $S = [Pb^{\cdot\cdot}].[OH']^2$ with concentrations expressed in gram-equivalents per litre, the formula:

$$\frac{RT}{2} \log_{10} S = -0{\cdot}669 + 0{\cdot}273 = -0{\cdot}396,$$

or $S = 2{\cdot}2 \times 10^{-14}$; taking into account[1] the degree of dissociation of the *n* alkali as 72 per cent., S becomes $1{\cdot}1 \times 10^{-14}$. A similar value will be obtained from the 0·01 *n* sodium hydroxide saturated by lead oxide, since the shift of the deposition potentials is from $-0{\cdot}669$ to $-0{\cdot}562 = 0{\cdot}107$ volt, which is very nearly $RT \log_{10} 100 = 2 \times 0{\cdot}058$ volt. We observe further that to a tenfold dilution of the lead content in the same alkaline solution corresponds a shift of about 28 millivolts, as would be expected from the equation of the ionic equilibrium

$$Pb^{\cdot\cdot} + 3\,OH' \rightleftarrows Pb(OH)'_3.$$

That only such monovalent plumbite ions are formed, in other words that "plumbous acid" is monobasic, is evident, *e.g.* from the shift of potentials between the normal solution with 3×10^{-4} gr.-mol. PbO content and the decinormal solution with 5×10^{-4} of PbO. The shift calculated for monovalent anions should be 92 millivolts, and for the dibasic plumbite anions, $Pb(OH)_4''$, it would be 29 millivolts larger still; the experiments show $-0{\cdot}726 + 0{\cdot}634 = 0{\cdot}092$, indicating that "plumbous acid" is monobasic.

The constant of the above reaction, which characterises the acidity of plumbous oxide can easily be calculated from any of these data; it is disregarding the degree of ionisation, $K_A = \frac{[Pb(OH)_3']}{[Pb^{\cdot\cdot}].[OH']^3} = 1{\cdot}4 \times 10^{12}$ and holds for any dilution. The more usual "acidic solubility product," which holds only for saturated plumbites, becomes then

$$[H^{\cdot}].[Pb(OH)_3'] = 3 \times 10^{-16}.$$

A similar research on zincates, done by the same student, reveals analogous relationships: from a normal sodium hydroxide saturated by metallic zinc dissolution the metal deposits at $-1{\cdot}330$ volt, whereas for a solution normal in zinc ions at $-0{\cdot}874$; the difference leads to the solubility product $= 1 \times 10^{16}$. The "zincic acid" appears also to be monobasic, at least towards normal or more dilute alkalies, its acidic constant K_A being $2{\cdot}3 \times 10^{14}$ and the "acidic solubility product" $4{\cdot}3 \times 10^{-16}$. The zinc oxide is therefore about as equally acidic as the plumbous oxide.

It may be mentioned that it has been found impossible to obtain these relations from zinc rod potentials, since the metal becomes passive in dilute alkali.

Thus this method may be suitably applied for the study of metallic complexes, such as amphoteric or acidic hydroxides, complex halides, amines, oxalates, etc.; the available range of potentials extends from 0 to $-2{\cdot}0$ volts from the calomel electrode.

Qualitative Electro-analysis.—An important feature observable on the curves in Fig. 3 is the turn to the horizontal in solutions with a very small amount of the metal, down from 10^{-4} gram-ions per litre. Evidently the current of about 10^{-7} amp., depositing every second at the cathode 10^{-12} gram-ions from the few cubic mm. in the immediate neighbourhood of the

[1] Glasstone, *J. Chem. Soc.*, 119 (1921), p. 1924, deduced from E.M.F. measurements $S = 1{\cdot}17 \times 10^{-15}$ in gram-ions per litre.

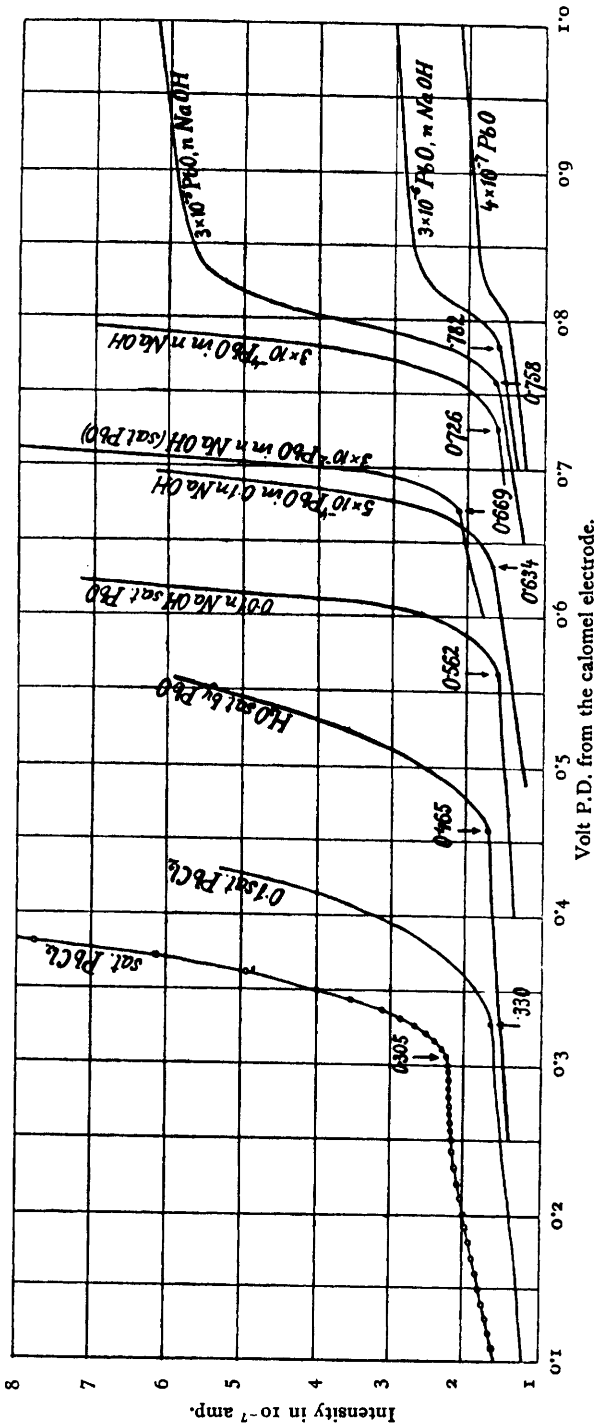
3×10^{-3} PbO, n NaOH
3×10^{-6} PbO, n NaOH
4×10^{-7} PbO
3×10^{-4} PbO in n NaOH
3×10^{-2} PbO in n NaOH (sat. PbO)
5×10^{-4} PbO in 0·1 n NaOH
0·01 n NaOH sat. PbO
H_2O sat. by PbO
0·1 sat. $PbCl_2$
sat. $PbCl_2$
·782
0·758
0·726
0·669
0·634
0·562
0·465
·330
0·305
Intensity in 10^{-7} amp.
Volt P.D. from the calomel electrode.

FIG. 3.

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dropping cathode, which contain at this dilution some 10^{-10} gram-ions of the metal, must soon exhaust the space round the drop, and the current intensity then depends chiefly on the amount diffused to. The metal in solution reveals itself in the form of a " wave " on the current-voltage curve quite distinctly when present in 10^{-6} gram-mols per litre, and even smaller amounts are still detectable.

The " wave " due to an admixture of 3×10^{-5} mols PbO per litre of a normal sodium hydroxide solution begins at $-0{\cdot}758v.$, and is about ten times more prominent than the " wave " due to a 3×10^{-6} PbO impurity, which rises at about $-0{\cdot}782$ volts from the calomel electrode. The latter " wave " again is much more distinct than that due to 4×10^{-7} gr.-mm. PbO. The same thing is observable in the case of dilute zincates in alkalis; here, of course, the potential of polarisation at which the wave due to zinc deposition begins is much more negative than in the case of lead.

It is obvious that by the determination of the position and size of such waves on the " current-voltage " curves traces of some metallic impurities in solutions can be identified; in other words, that this method can serve as a means of *qualitative electro-analysis.* Some instances of this are graphically represented in Fig. 4.

The curve (1)Zn was obtained from electrolysis of a normal zinc chloride solution, which was prepared by the solution of the equivalent amount of pure (Merck's " pro analysi ") zinc in normal hydrochloric acid, so that no metallic residue was left. Two big waves appear (at $-0{\cdot}41$ and $-0{\cdot}59$ volt), which bring the galvanometer deflexion off the scale before the actual deposition of zinc ions in mercury can be reached. The proper value of the zinc deposition potential is obtained only when an excess of the metal is treated with hydrochloric acid, and the presence of the metallic phase does not allow impurities of a more noble character than that of zinc to enter the solution. In this way the curve (2)Zn was obtained. That $-0{\cdot}874$ is the true value of zinc deposition, and that it does not relate to another substance, is ascertained on subsequent dilutions, when only a regular shift (by about 28 millivolts per tenfold dilution) is observable, until the curves turn into the form of " waves " at about 10^{-5} gr.-mols of Zn per litre. Such extreme dilutions have to be made up with a conducting solution, *e.g.* in a pure $0{\cdot}1n$ potassium chloride, which does not interfere with the deposition of zinc. If the resistance of the solutions is large, the slope of the ascending branch of the curve is lessened and the turning-point becomes indistinct.

A tenfold dilution of the solution (1)Zn causes a great diminution of the two waves, so that the determination of the zinc deposition at $-0{\cdot}900v$ appears within the galvanometer scale.

Evidently the very pure zinc contains at least two impurities of nobler metals in an amount of about 1 part in 100,000, *i.e.* 0·001 per cent. Judging from their potential positions, they are probably indium and lead. The first wave (at $-0{\cdot}41$) appears near the cadmium normal deposition potential; however, sulphureted hydrogen produced only a faint yellow coloration in this solution, showing that there is so little of cadmium present that its deposition could not begin before $-0{\cdot}50$ volts. It has been ascertained later on, that the first wave is due to lead and the second to indium impurities.

The next curve shows the electrolysis of a ferrous solution with an addition of 5×10^{-6} mols of zinc chloride per litre. Zinc, of course, deposits before iron, behaving at the mercury electrode as a more noble metal.

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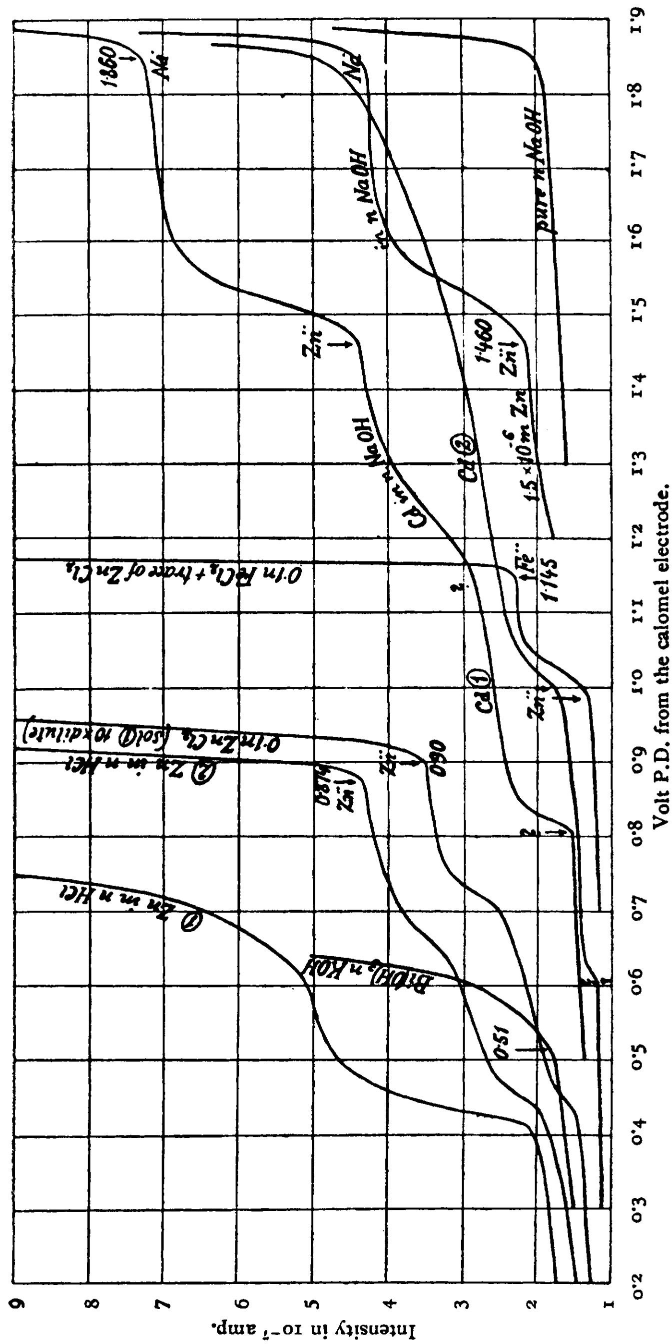

Fig. 4.

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The curve denoted "Cd(1)" shows the polarisation current obtained in a normal sodium hydroxide solution, which was allowed to act for three months on a thin strip of pure (Kahlbaum's "pro analysi") cadmium. Although cadmium apparently does not dissolve in alkaline solutions, yet it gives rise after some weeks to a precipitate of cadmium hydroxide, and the metallic surface turns black and seems corroded. The alkaline extract contains zinc, as it is evident from the large wave coinciding in size and position with an artificial zinc addition to normal sodium hydroxide (see curve below "Cd(2)"). The alkaline extract when filtered and neutralised by pure hydrochloric acid gave the curve denoted "Cd(2)." In this the zinc impurity is found to deposit in the expected place at $-$ 1·00*v*. (it has been also ascertained as zinc micro-analytically). The other impurity, producing a wave at $-$ 0·810*v*. in the alkaline and at $-$ 0·61*v*. in the neutralised solution, must be due to an amphoteric substance, perhaps cadmium hydroxide itself, if it possesses a faintly acidic character.

The more markedly pronounced acidity of bismuth hydroxide, which manifests itself in dissolving in very concentrated alkalies [1] is easily shown from the polarisation curve of normal sodium hydroxide to which some bismuth hydroxide has been added; the increase of current due to bismuth deposition starts near $-$ 0·5 volt.

These examples suggest the possibility of working out a systematic method of analysis by means of the dropping cathode.

Conclusions.

The experimental results given above show that a polarised drop of a mercury capillary cathode represents a reversible state of equilibrium. On each polarised drop the number of ions is instantly deposited sufficient to charge it by their solution tension to the balancing back E.M.F. The almost streamless condition of polarisation excludes the secondary effects of the current so far that the potentials at which ions are deposited from various concentrations change like the potentials of concentration cells with reversible metallic electrodes. Thus this arrangement allows an extension to the study of tri- or tetra-valent ionic concentration cells. Further, since the freshly deposited atoms are always in an active condition, the conclusion seems justified, that any retardation phenomena observed in the deposition cannot be due to surface conditions, but must be rather sought in an imperfect ionic equilibrium of the solution (*e.g.* in the case of arsenic).

This circumstance enables us also to decide whether some metallic compounds exist as a true or as a colloidal solution.

The applicability of this method is limited on one side by the most 'unnoble' lithium deposition-potential at about $-$ 2·1 volts from the normal calomel electrode, on the other it stretches to + 0·2 volt, where the oxidation of mercury begins.

Finally it may be pointed out that the method can be equally adapted for the study of non-aqueous solutions (see Dr. Shikata's subsequent communication).

[1] *E.g.* Grube and Schweigardt, *z. f. Elektrochemie*, **29** (1923), p. 257.

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